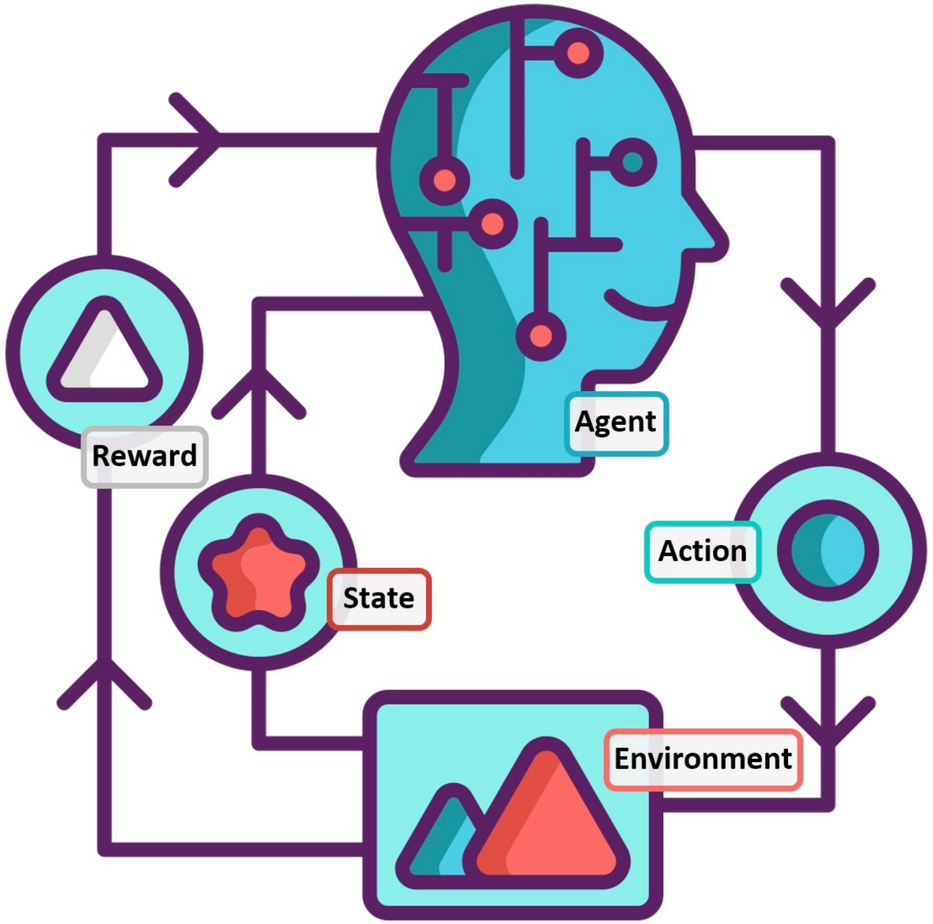
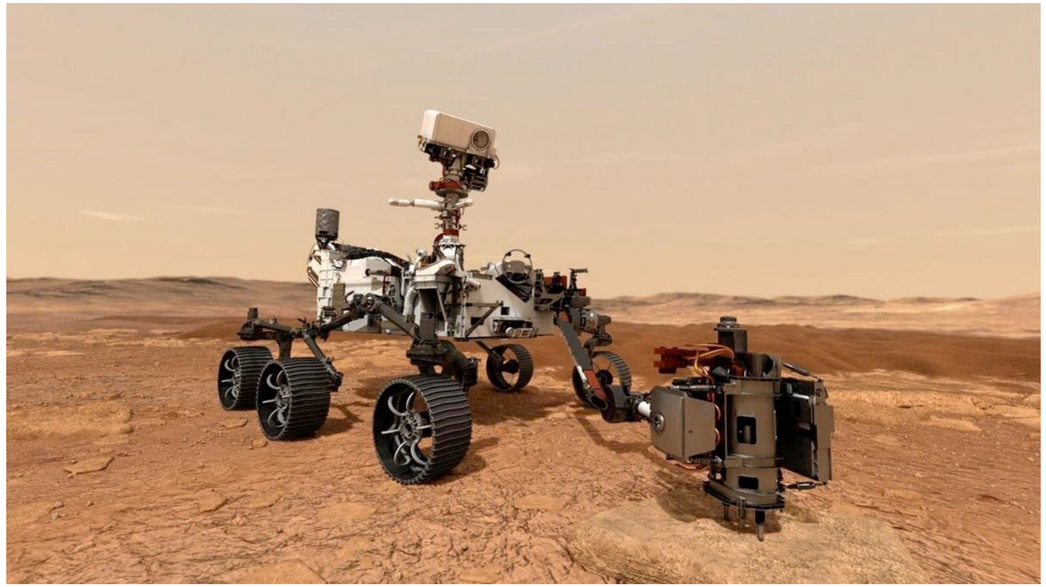
Algunos hitos del DRL

Manuel Sánchez-Montañés

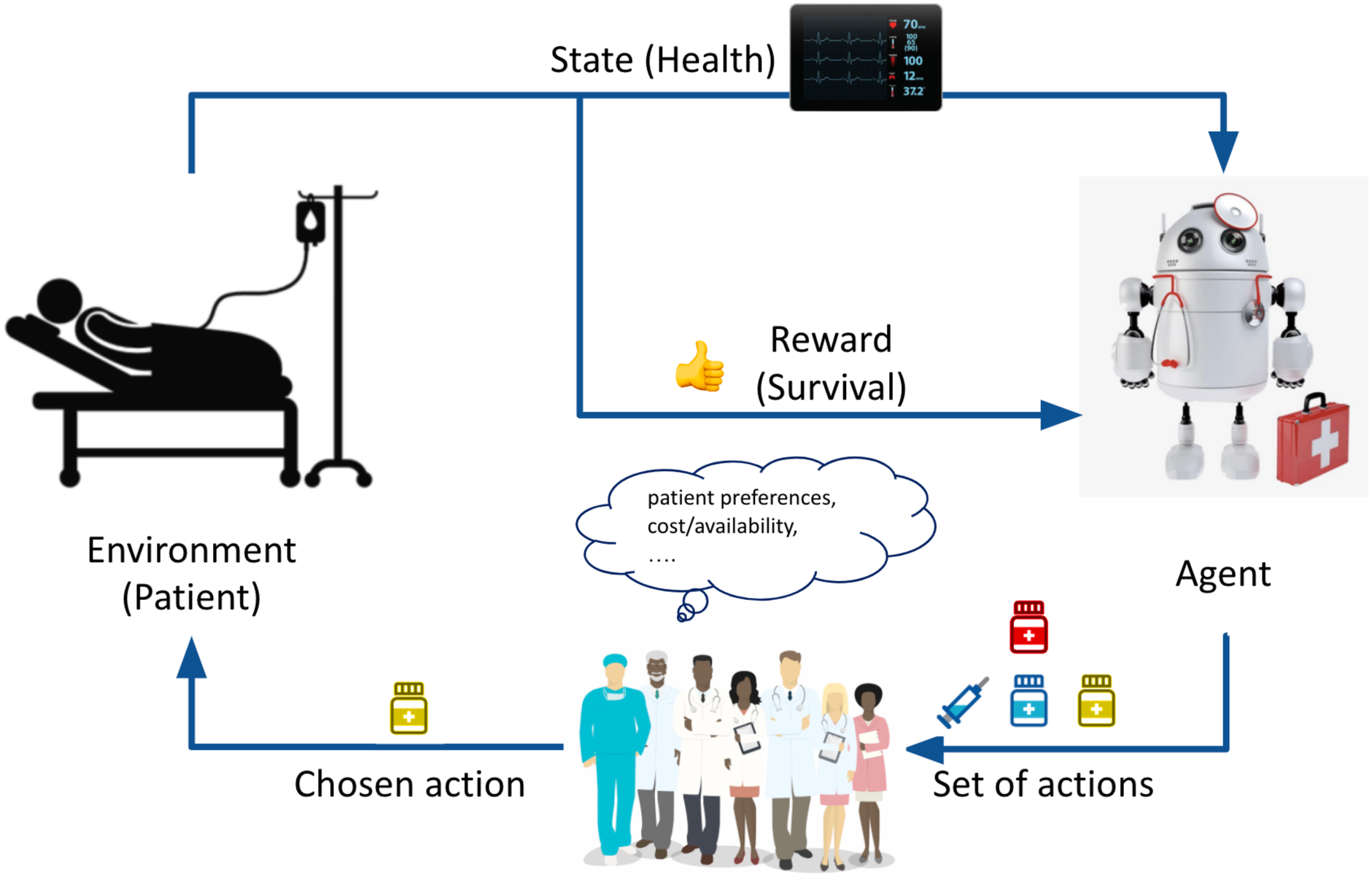

Reward
Agent
State
Action
Environment







https://www.youtube.com/watch?v=paombvhdXGg


State (Health)
Reward
(Survival)
Environment
(Patient)
patient preferences,
cost/availability,
….
Agent
Chosen action
Set of actions

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journal homepage: www.elsevier.com/locate/eswa

A reinforcement learning approach for individualizing erythropoietin dosages in hemodialysis patients

José D. Martín-Guerrero [a,*], Faustino Gomez [b], Emilio Soria-Olivas [a], Jürgen Schmidhuber [b], Mónica Climente-Martí [c], N. Víctor Jiménez-Torres [c,d]

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[b] Istituto Dalle Molle di Studi sull'Intelligenza Artificiale (IDSIA), Lugano, Switzerland
[c] Pharmacy Unit, University Hospital, Dr. Peset, Valencia, Spain
[d] Pharmacy and Pharmaceutical Technology Department, University of Valencia, Spain

Nine actions were used for both methods:

1: Increase EPO β dosage.
2: Decrease EPO β dosage.
3: Increase darbepoietin α dosage.
4: Decrease darbepoietin α dosage.
5: Increase no. of EPO β administrations.
6: Decrease no. of EPO β administrations.
7: Increase no. of darbepoietin α administrations.
8: Decrease no. of darbepoietin α administrations.
9: No change in dosage or number of administrations.

$$\text{reward} = (\text{reward})_{\text{Hb}} + (\text{reward})_{\text{change}} \tag{6}$$

$$(\text{reward})_{\text{Hb}} = \begin{cases} 10, & \text{if Hb} \in [11.5 - 12.5]\ \text{g/dl} \\ 0, & \text{if Hb} \in [10.0 - 11.5)\ \text{g/dl} \\ & \text{or Hb} \in (12.5 - 13.0)\ \text{g/dl} \\ -5, & \text{if Hb} \in [9.0 - 10.0)\ \text{g/dl} \\ & \text{or Hb} \in [13.0 - 14.0)\ \text{g/dl} \\ -10, & \text{otherwise} \end{cases} \tag{7}$$

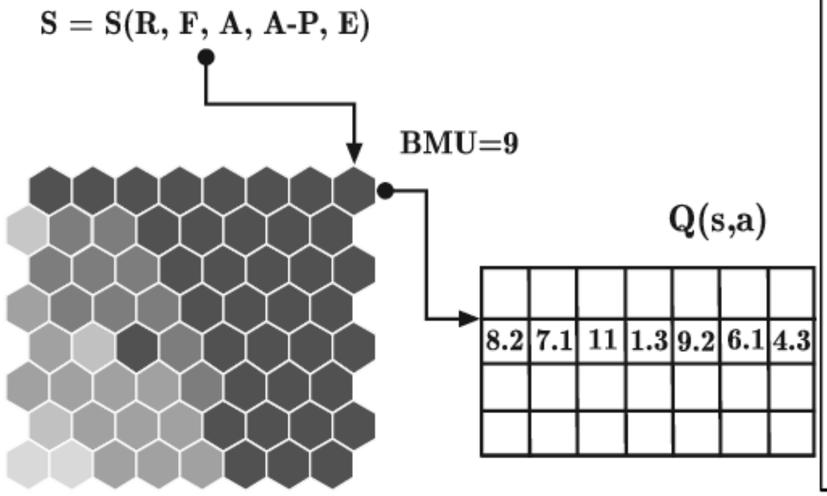
$$(\text{reward})_{\text{change}} = \begin{cases} 0, & \text{if } |\text{Hb}(t) - \text{Hb}(t-1)| < 1\ \text{g/dl} \\ -5, & \text{if } 1 \leqslant |\text{Hb}(t) - \text{Hb}(t-1)| < 1.5\ \text{g/dl} \\ -10, & \text{otherwise} \end{cases} \tag{8}$$

Expert Systems with Applications
Volume 36, Issue 4, May 2009, Pages 8022-8031

Assigning discounts in a marketing campaign by using reinforcement learning and neural networks

Gabriel Gómez-Pérez [a], José D. Martín-Guerrero [a], Emilio Soria-Olivas [a], Emili Balaguer-Ballester [b], Alberto Palomares [b], Nicolás Casariego [c]

IDAL
Intelligent Data Analysis Laboratory



RL Model

1) *State.* $s_t = BMU_i$, *for the i-th customer transaction.*
2) *Action.* $a_t = a_t(V_C)$ *where* V_C *stands for the virtual credits assigned.*
3) *Immediate reward.* $r_t = LTV$.

Algorithm procedure

1) *Training a SOM.*
2) *The state of customers is computed through their BMU at time t.*
3) *Action carried out by the company.*
4) *Once reward is obtained, the following state is obtained.*
5) *Computation of* $Q^{\pi}(s, a)$.
6) *Estimation of the optimal policy* $\pi^*(s)$.

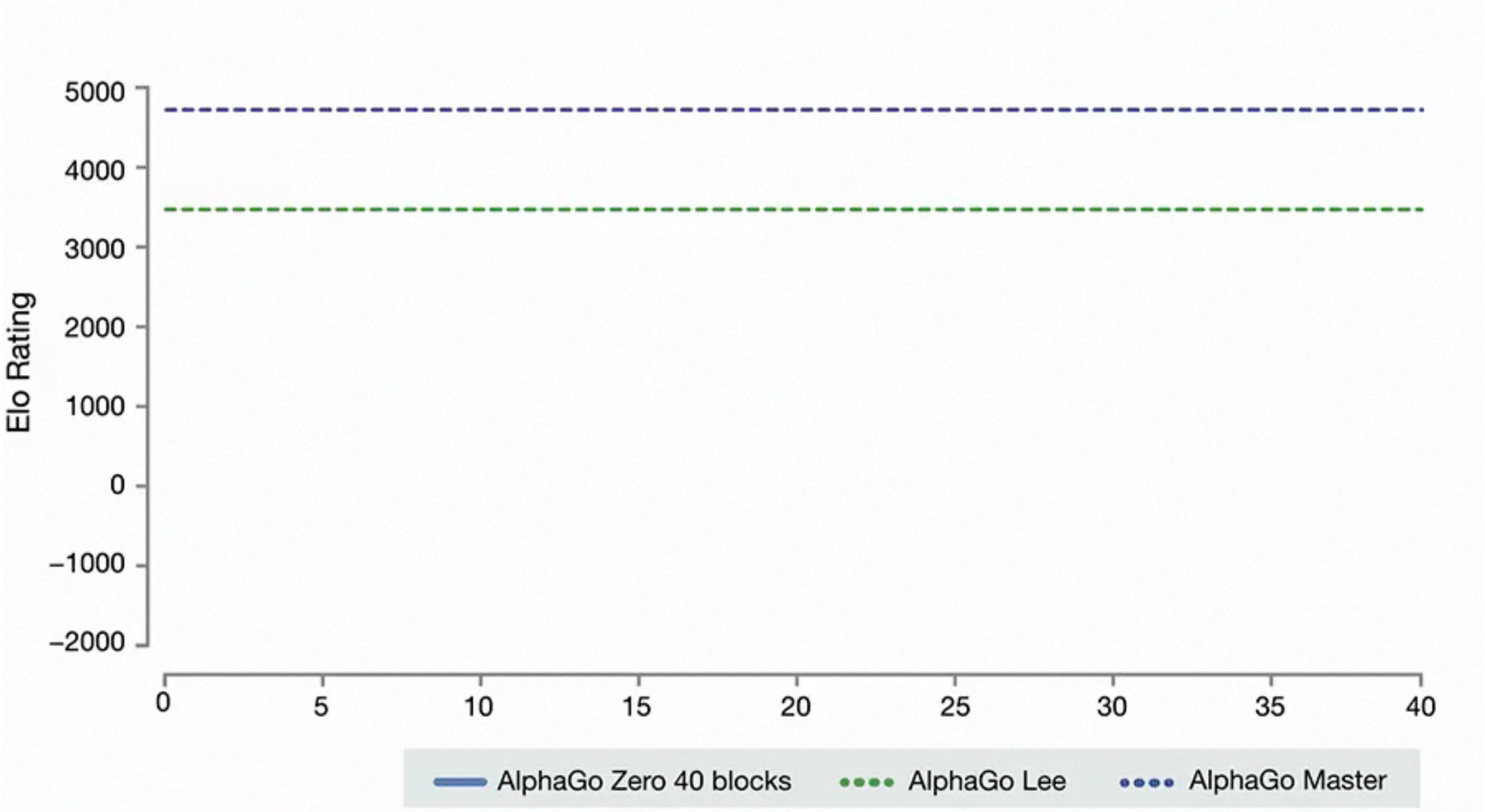
...s of state aggregation using the self-organizing map (SOM). The Q-table is filled in with the values of the best matching units (BMU...



Google DeepMind


KBA
Korea Baduk Association
Google DeepMind
Challenge Match
8 - 15 March 2016
AlphaGo
ALPHAGO
00:12:30
LEE SEDOL
00:34:31
AlphaGo
Lee Sedol

Aprendizaje reforzado



Research

Discovering novel algorithms with AlphaTensor

October 5, 2022



$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} & a_{1,5} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} & a_{2,5} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} & a_{3,5} \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} & a_{4,5} \end{bmatrix} \times \begin{bmatrix} b_{1,1} & b_{1,2} & b_{1,3} & b_{1,4} & b_{1,5} \\ b_{2,1} & b_{2,2} & b_{2,3} & b_{2,4} & b_{2,5} \\ b_{3,1} & b_{3,2} & b_{3,3} & b_{3,4} & b_{3,5} \\ b_{4,1} & b_{4,2} & b_{4,3} & b_{4,4} & b_{4,5} \\ b_{5,1} & b_{5,2} & b_{5,3} & b_{5,4} & b_{5,5} \end{bmatrix} = \begin{bmatrix} c_{1,1} & c_{1,2} & c_{1,3} & c_{1,4} & c_{1,5} \\ c_{2,1} & c_{2,2} & c_{2,3} & c_{2,4} & c_{2,5} \\ c_{3,1} & c_{3,2} & c_{3,3} & c_{3,4} & c_{3,5} \\ c_{4,1} & c_{4,2} & c_{4,3} & c_{4,4} & c_{4,5} \end{bmatrix}$$

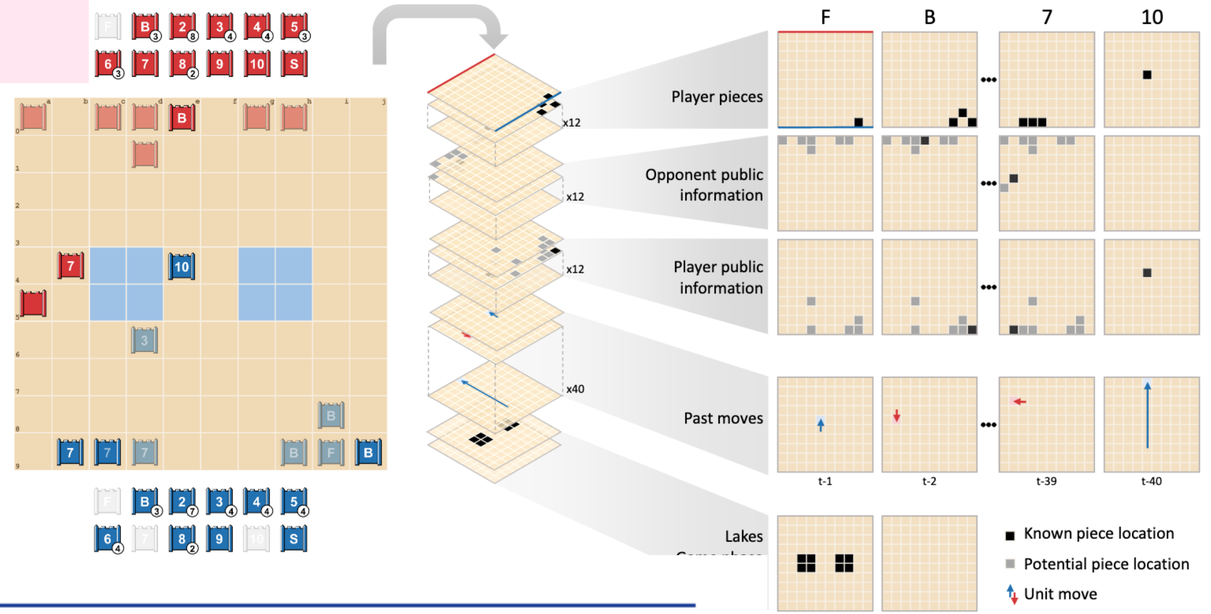
$$\begin{aligned}
h_{1} &= a_{3,2}\left(-b_{2,1} - b_{2,5} - b_{3,1}\right) \\
h_{2} &= \left(a_{2,2} + a_{2,5} - a_{3,5}\right)\left(-b_{2,5} - b_{5,1}\right) \\
h_{3} &= \left(-a_{3,1} - a_{4,1} + a_{4,2}\right)\left(-b_{1,1} + b_{2,5}\right) \\
h_{4} &= \left(a_{1,2} + a_{1,4} + a_{3,4}\right)\left(-b_{2,5} - b_{4,1}\right) \\
h_{5} &= \left(a_{1,5} + a_{2,2} + a_{2,5}\right)\left(-b_{2,4} + b_{5,1}\right) \\
h_{6} &= \left(-a_{2,2} - a_{2,5} - a_{4,5}\right)\left(b_{2,3} + b_{5,1}\right) \\
h_{7} &= \left(-a_{1,1} + a_{4,1} - a_{4,2}\right)\left(b_{1,1} + b_{2,4}\right) \\
h_{8} &= \left(a_{3,2} - a_{3,3} - a_{4,3}\right)\left(-b_{2,3} + b_{3,1}\right) \\
h_{9} &= \left(-a_{1,2} - a_{1,4} + a_{4,4}\right)\left(b_{2,3} + b_{4,1}\right) \\
h_{10} &= \left(a_{2,2} + a_{2,5}\right) b_{5,1} \\
h_{11} &= \left(-a_{2,1} - a_{4,1} + a_{4,2}\right)\left(-b_{1,1} + b_{2,2}\right) \\
h_{12} &= \left(a_{4,1} - a_{4,2}\right) b_{1,1} \\
h_{13} &= \left(a_{1,2} + a_{1,4} + a_{2,4}\right)\left(b_{2,2} + b_{4,1}\right) \\
h_{14} &= \left(a_{1,3} - a_{3,2} + a_{3,3}\right)\left(b_{2,4} + b_{3,1}\right) \\
h_{15} &= \left(-a_{1,2} - a_{1,4}\right) b_{4,1} \\
h_{16} &= \left(-a_{3,2} + a_{3,3}\right) b_{3,1} \\
h_{17} &= \left(a_{1,2} + a_{1,4} - a_{2,1} + a_{2,2} - a_{2,3} + a_{2,4} - a_{3,2} + a_{3,3} - a_{4,1} + a_{4,2}\right) b_{2,2}
\end{aligned}$$

$$\begin{aligned}
h_{51} &= a_{2,2}\left(b_{2,1} + b_{2,2} - b_{5,1}\right) \\
h_{52} &= a_{4,2}\left(b_{1,1} + b_{2,1} + b_{2,3}\right) \\
h_{53} &= -a_{1,2}\left(-b_{2,1} + b_{2,4} + b_{4,1}\right) \\
h_{54} &= \left(a_{1,2} + a_{1,4} - a_{2,2} - a_{2,5} - a_{3,2} + a_{3,3} - a_{4,2} + a_{4,3} - a_{4,4} - a_{4,5}\right) b_{2,3} \\
h_{55} &= \left(a_{1,4} - a_{4,4}\right)\left(-b_{2,3} + b_{3,1} + b_{3,3} - b_{3,4} + b_{4,3} - b_{4,4}\right) \\
h_{56} &= \left(a_{1,1} - a_{1,5} - a_{4,1} + a_{4,5}\right)\left(b_{3,1} + b_{3,3} - b_{3,4} + b_{5,1} + b_{5,3} - b_{5,4}\right) \\
h_{57} &= \left(-a_{3,1} - a_{4,1}\right)\left(-b_{1,3} - b_{1,5} - b_{2,5} - b_{5,1} - b_{5,3} - b_{5,5}\right) \\
h_{58} &= \left(-a_{1,4} - a_{1,5} - a_{3,4} - a_{3,5}\right)\left(-b_{5,1} + b_{5,4} - b_{5,5}\right) \\
h_{59} &= \left(-a_{3,3} + a_{3,4} - a_{4,3} + a_{4,4}\right)\left(b_{4,1} + b_{4,3} + b_{4,5} + b_{5,1} + b_{5,3} + b_{5,5}\right) \\
h_{60} &= \left(a_{2,5} + a_{4,5}\right)\left(b_{2,3} - b_{3,1} - b_{3,2} - b_{3,3} - b_{5,2} - b_{5,3}\right) \\
h_{61} &= \left(a_{1,4} + a_{3,4}\right)\left(b_{1,1} - b_{1,4} + b_{1,5} - b_{2,5} - b_{4,4} + b_{4,5} - b_{5,1} + b_{5,4} - b_{5,5}\right) \\
h_{62} &= \left(a_{2,1} + a_{4,1}\right)\left(b_{1,2} + b_{1,3} + b_{2,2} - b_{4,1} - b_{4,2} - b_{4,3}\right) \\
h_{63} &= \left(-a_{3,3} - a_{4,3}\right)\left(-b_{2,3} - b_{3,3} - b_{3,5} - b_{4,1} - b_{4,3} - b_{4,5}\right) \\
h_{64} &= \left(a_{1,1} - a_{1,3} - a_{1,4} + a_{3,1} - a_{3,3} - a_{3,4}\right)\left(b_{1,1} - b_{1,4} + b_{1,5}\right) \\
h_{65} &= \left(-a_{1,1} + a_{4,1}\right)\left(-b_{1,3} + b_{1,4} + b_{2,4} - b_{5,1} - b_{5,3} + b_{5,4}\right) \\
h_{66} &= \left(a_{1,1} - a_{1,2} + a_{1,3} - a_{1,5} - a_{2,2} - a_{2,5} - a_{3,2} + a_{3,3} - a_{4,1} + a_{4,2}\right) b_{2,4} \\
h_{67} &= \left(a_{2,5} - a_{3,5}\right)\left(b_{1,1} + b_{1,2} + b_{1,5} - b_{2,5} - b_{4,1} - b_{4,2} - b_{4,5} + b_{5,2} + b_{5,5}\right)
\end{aligned}$$

Research

Mastering Stratego, the classic game of imperfect information

December 1, 2022



	Chess	Poker	Go 19x19	Stratego
Turns in a game	~60	~15	~300	~1000
Starting configurations	1	10^{6}	1	10^{66}
Game tree complexity	10^{123}	10^{17}	10^{360}	10^{535}

The scale of the differences between chess, poker, Go, and Stratego.

Article | Open Access | Published: 07 June 2023

Faster sorting algorithms discovered using deep reinforcement learning

Daniel J. Mankowitz, Andrea Michi, Anton Zhernov, Marco Gelmi, Marco Selvi, Cosmin Paduraru, Edouard Leurent, Shariq Iqbal, Jean-Baptiste Lespiau, Alex Ahern, Thomas Köppe, Kevin Millikin, Stephen Gaffney, Sophie Elster, Jackson Broshear, Chris Gamble, Kieran Milan, Robert Tung, Minjae Hwang, Taylan Cemgil, Mohammadamin Barekatain, Yujia Li, Amol Mandhane, Thomas Hubert, ... David Silver + Show authors

Nature **618**, 257–263 (2023) | Cite this article

https://www.deepmind.com/blog/alphadev-discovers-faster-sorting-algorithms

Original

```
Memory[0] = A
Memory[1] = B
Memory[2] = C

mov Memory[0] P  // P = A
mov Memory[1] Q  // Q = B
mov Memory[2] R  // R = C

mov R S
cmp P R
cmovg P R  // R = max(A, C)
cmovl P S  // S = min(A, C)
mov S P    // P = min(A, C)
cmp S Q
cmovg Q P // P = min(A, B, C)
cmovg S Q // Q = max(min(A, C), B)

mov P Memory[0]  // = min(A, B, C)
mov Q Memory[1]  // = max(min(A, C), B)
mov R Memory[2]  // = max(A, C)
```

AlphaDev

```
Memory[0] = A
Memory[1] = B
Memory[2] = C

mov Memory[0] P  // P = A
mov Memory[1] Q  // Q = B
mov Memory[2] R  // R = C

mov R S
cmp P R
cmovg P R  // R = max(A, C)
cmovl P S  // S = min(A, C)

cmp S Q
cmovg Q P  // P = min(A, B)
cmovg S Q  // Q = max(min(A, C), B)

mov P Memory[0]  // = min(A, B)
mov Q Memory[1]  // = max(min(A, C), B)
mov R Memory[2]  // = max(A, C)
```